Supplementary material

Beyond Point Accuracy and Rule Compliance: Layered Evaluation of a Constrained Neuro-Symbolic Decision-Support System

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Table S1 Pooled action confusion matrices for the integrated backbone (B4) and the full method (P) across three seeds, 57 seed-case outputs on 19 unique cases

System	Reference action	Predicted Increase	Predicted Maintain	Predicted Reduce	Predicted Investigate	Row total
B4	Increase	8	1	0	0	9
	Maintain	0	16	0	5	21
	Reduce	0	0	15	0	15
	Investigate	0	0	6	6	12
	Total	8	17	21	11	57
P	Increase	8	1	0	0	9
	Maintain	0	18	0	3	21
	Reduce	0	0	9	6	15
	Investigate	0	0	0	12	12
	Total	8	19	9	21	57

The 57 outputs are repeated runs of three seeds over the same 19 cases, not 57 independent observations. The generalisation unit is the campaign case.

Table S2 Seed-level preferred-action accuracy, Macro F1, and Top-2 agreement for every neural configuration on the original 19-case cohort

Configuration	Seed	Accuracy (%)	Macro F1	Top-2 (%)
Proposed System - Full Framework	42	78.95	0.7836	89.47
	52	84.21	0.8501	89.47
	62	84.21	0.8501	89.47
Unconstrained AI Baseline	42	89.47	0.8865	100.00
	52	94.74	0.9530	100.00
	62	94.74	0.9530	100.00
Branch Ablation - CampaignNet Only	42	36.84	0.3237	47.37
	52	47.37	0.3927	57.89
	62	47.37	0.3927	52.63
Branch Ablation - ActionNet Only	42	89.47	0.8865	100.00
	52	94.74	0.9530	100.00
	62	89.47	0.8865	100.00
Branch Ablation - Fused CampaignNet-ActionNet	42	89.47	0.8865	100.00
	52	94.74	0.9530	100.00
	62	94.74	0.9530	100.00
Proposed Ablation - Without Intent	42	68.42	0.6953	84.21
	52	78.95	0.7974	89.47
	62	73.68	0.7614	89.47
Proposed Ablation - Without Rules	42	89.47	0.8865	100.00
	52	94.74	0.9530	100.00
	62	94.74	0.9530	100.00
Proposed Ablation - Without Calibration	42	78.95	0.7836	89.47
	52	84.21	0.8501	89.47

Configuration	Seed	Accuracy (%)	Macro F1	Top-2 (%)
	62	84.21	0.8501	89.47
Proposed Ablation - Without Structured Explanation	42	78.95	0.7836	89.47
	52	84.21	0.8501	89.47
	62	84.21	0.8501	89.47
Proposed Ablation - Without Review Gate	42	78.95	0.7836	89.47
	52	84.21	0.8501	89.47
	62	84.21	0.8501	89.47
B4: Integrated Learned Backbone	42	73.68	0.7256	94.74
	52	84.21	0.8320	100.00
	62	78.95	0.7917	89.47

Accuracy and Top-2 agreement are percentages of the 19 unique cases; Macro F1 is the unweighted mean over the four action classes. Deterministic comparators are evaluated once and are reported in Table 1 of the main text.

Table S3 Reliability-bin composition for the full method under raw and temperature-scaled probabilities, seed 42, validation and test partitions

Partition	Probabilities	Bin	Interval	n	k	Mean confidence	Observed accuracy	Absolute gap	ECE contribution
Validation	Raw	1	[0.0, 0.2)	0	—	—	—	—	0.0000
Validation	Raw	2	[0.2, 0.4)	1	1	0.3856	1.0000	0.6144	0.0323
Validation	Raw	3	[0.4, 0.6)	16	10	0.4855	0.6250	0.1395	0.1175
Validation	Raw	4	[0.6, 0.8)	0	—	—	—	—	0.0000
Validation	Raw	5	[0.8, 1.0]	2	1	1.0000	0.5000	0.5000	0.0526
Validation	Calibrated	1	[0.0, 0.2)	0	—	—	—	—	0.0000
Validation	Calibrated	2	[0.2, 0.4)	17	11	0.3613	0.6471	0.2858	0.2557
Validation	Calibrated	3	[0.4, 0.6)	0	—	—	—	—	0.0000
Validation	Calibrated	4	[0.6, 0.8)	0	—	—	—	—	0.0000
Validation	Calibrated	5	[0.8, 1.0]	2	1	1.0000	0.5000	0.5000	0.0526
Test	Raw	1	[0.0, 0.2)	0	—	—	—	—	0.0000
Test	Raw	2	[0.2, 0.4)	0	—	—	—	—	0.0000
Test	Raw	3	[0.4, 0.6)	18	14	0.5018	0.7778	0.2760	0.2615
Test	Raw	4	[0.6, 0.8)	0	—	—	—	—	0.0000
Test	Raw	5	[0.8, 1.0]	1	1	1.0000	1.0000	0.0000	0.0000
Test	Calibrated	1	[0.0, 0.2)	0	—	—	—	—	0.0000
Test	Calibrated	2	[0.2, 0.4)	16	14	0.3644	0.8750	0.5106	0.4300
Test	Calibrated	3	[0.4, 0.6)	2	0	0.5091	0.0000	0.5091	0.0536
Test	Calibrated	4	[0.6, 0.8)	0	—	—	—	—	0.0000
Test	Calibrated	5	[0.8, 1.0]	1	1	1.0000	1.0000	0.0000	0.0000

n is the number of the 19 cases falling in the bin and k the number of those cases whose preferred action was recovered. Only one or two bins are occupied in each partition, which is why a 19-case cohort cannot support a stable bin-level calibration claim.

Table S4 Review-state composition for the original 19-case seed-42 audit

Review state or derived quantity	Count	Denominator	Rate (%)
RECOMMEND	9	19	47.37
CHECK ADVISED	1	19	5.26
HUMAN REVIEW REQUIRED	8	19	42.11
NO SAFE RECOMMENDATION	1	19	5.26
Automatic coverage	9	19	47.37
Review demand (non-automatic)	10	19	52.63

Single-seed audit. One case fell outside the training benchmark envelope and is counted within the non-automatic states. These are policy diagnostics, not seed-averaged estimates.